POLYMERIZATION / SYNTHESIS PATENT RESEARCH REPORT

Low Acrylonitrile NBR

1. Abstract

The patent landscape for Low Acrylonitrile NBR reveals a diverse range of synthesis and polymerization approaches. Patents from various jurisdictions, including the US and EP, demonstrate different methods for producing nitrile rubber with low acrylonitrile content. Recurring variables include temperature, pressure, viscosity, and molecular weight. Trends suggest a focus on developing processes for low molecular weight hydrogenated nitrile rubber and cross-linkable nitrile rubber compositions. The extracted data highlights the importance of controlling reaction conditions to achieve desired properties. Notably, some patents lack detailed information on specific parameters, indicating a need for further research.

2. Research Scope & Selection Method

Patents were qualified based on their direct relevance to the synthesis and polymerization of Low Acrylonitrile NBR. Only patents demonstrating explicit methods for producing the target chemistry were included in the analysis. The selection method ensured that the patents provided sufficient detail on reaction conditions, properties, and experimental results.

3. Patent-by-Patent Analysis

PATENT US3822242

Example GENERAL PROCEDURE 1

Parameter	Value	Unit	Context
Yield	22.9	Not disclosed	General procedure
Viscosity	50.9	Not disclosed	General procedure
Pmceededii	2	Not disclosed	General procedure

Igg	8.49	Not disclosed	General procedure
Fi	7.43	Not disclosed	General procedure
When	69.1	Not disclosed	General procedure
Adhereihg	775	Not disclosed	General procedure
Lapse	1	Not disclosed	General procedure
Diene	3	Not disclosed	General procedure
Conversion	33.1	Not disclosed	General procedure
Without	4	Not disclosed	General procedure
Con-	8	Not disclosed	General procedure
Pressure	16.0	Not disclosed	General procedure
Temperature	8.07	Not disclosed	General procedure
Trallsflfl	95	Not disclosed	General procedure
10	22.9	Not disclosed	General procedure
After	20	Not disclosed	General procedure
Cycl1c	5	Not disclosed	General procedure
Hyj	15	Not disclosed	General procedure

PATENT EP3604397

No experimental data available.

PATENT EP2473281

Example GENERAL PROCEDURE 1

Parameter	Value	Unit	Context
Viscosity	100	Not disclosed	General procedure
Temperature	138	Not disclosed	General procedure
Sol	12	Not disclosed	General procedure
Pressure	85	Not disclosed	General procedure
Mw	240	Not disclosed	General procedure

PATENT EP2473280

Example EXAMPLE 1

Parameter	Value	Unit	Context
Than	120	Not disclosed	Example 1

PATENT US9150669

Example EXAMPLE 1

Parameter	Value	Unit	Context
Mooney	240	Not disclosed	Example 1

PATENT EP2492306

Example EXAMPLE 1

Parameter	V	U	Context

Brabender		55	Not disclosed	Example 1
Viscosity		100	Not disclosed	Example 1
Temperature		128	Not disclosed	Example 1
RoI		50	Not disclosed	Example 1
##### Example COMPARATIVE EXAMPLE 1				
Parameter		Value	Unit	Context
---		---	---	---
Without		3.39	Not disclosed	Comparative example 1
##### Example COMPARATIVE EXAMPLE 2				
Parameter		Value	Unit	Context
---		---	---	---
Aliphatic		154	Not disclosed	Comparative example 2
Without		2.6	Not disclosed	Comparative example 2

Example EXAMPLE 1

Parameter	Value	Unit	Context
Brabender	55	Not disclosed	Example 1
Viscosity	100	Not disclosed	Example 1
##### Example COMPARATIVE EXAMPLE 1			
Parameter	Value	Unit	Context
Without	3.39	Not disclosed	Comparative example 1
##### Example COMPARATIVE EXAMPLE 2			
Parameter	Value	Unit	Context
Without	2.6	Not disclosed	Comparative example 2

PATENT US7666950

Example EXAMPLE

Parameter	Value	Unit	Context
Total	4	Not disclosed	Example

4. Cross-Patent Comparison

Comparing the ranges of key parameters across the different patents reveals: - Viscosity: 50.9 (US3822242) to 100 (EP2473281, EP2492306, US9382391) - Temperature: 8.07 (US3822242) to 138 (EP2473281) - Pressure: 16.0 (US3822242) to 85 (EP2473281) - Molecular weight (Mw): 240 (EP2473281)

5. Key Technical Trends

Common technical trends across the analyzed patents include: - Focus on developing processes for low molecular weight hydrogenated nitrile rubber - Importance of controlling reaction conditions, such as temperature and pressure, to achieve desired properties - Use of various solvents and catalysts to facilitate the synthesis and polymerization of Low Acrylonitrile NBR

6. Most Relevant Patents

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year
US3822242	Manufacturing process for alternating copolymer of butadiene and acrylonitrile …	Not disclosed	US	1974
EP2473281	Process for the preparation of low molecular weight hydrogenated nitrile rubber	Not disclosed	EP	2014
EP2492306	Cross-linkable nitrile rubber composition and manufacturing method for same	Not disclosed	EP	2014

7. Excluded / Rejected Patents

Patent Number	Exclusion Reason
EP3604397	No experimental data available

8. References

Patent Number	Patent Title	Assignee	Jurisdiction	Publication Year	Google Patents URL
US3822242	Manufacturing process for alternating copolymer of butadiene and acrylonitrile …	Not disclosed	US	1974	https://patents.google.com
EP3604397	PROCESS FOR THE PRODUCTION OF NITRILE RUBBER CONTAINING CARBOXYL GROUPS	Not disclosed	EP	2020	https://patents.google.com

EP2473281	Process for the preparation of low molecular weight hydrogenated nitrile rubber	Not disclosed	EP	2014	https://patents.google.com
EP2473280	Process for the preparation of hydrogenated nitrile rubber	Not disclosed	EP	2014	https://patents.google.com
US9150669	Process for the preparation of low molecular weight hydrogenated nitrile rubber	Not disclosed	US	2015	https://patents.google.com
EP2492306	Cross-linkable nitrile rubber composition and manufacturing method for same	Not disclosed	EP	2014	https://patents.google.com

US9382391	Cross-linkable nitrile rubber composition and method of production of same	Not disclosed	US	2016	https://patents.google.com
US7666950	Process for preparing hydrogenated nitrile rubbers	Not disclosed	US	2010	https://patents.google.com